GRAPHS AND MATRICES

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Contents

Abstract. This is a very brief note on **Spectral Graph Theory**.

1. Adjacency Matrices

Given a graph, one can associate various matrices to encode its information. The *adjacency matrix* A of a graph $X = (V, E)$ is the matrix whose rows and columns are indexed by the vertices of X, where $A(x, y)$ equals the number of edges between x and y. When necessary to indicate the dependence on X, we denote A by $A(X)$.

A number λ is called an *eigenvalue* of A if there exists a non-zero vector $u \in \mathbb{R}^n$ such that $Au = \lambda u$. This means that λ is a root of the *characteristic polynomial* $\det(tI_n - A)$.

If n denotes the number of vertices of X, then A is an $n \times n$ real and symmetric matrix, and therefore by linear algebra, it has n real eigenvalues, which we denote as follows: $\lambda_1 \geq \cdots \geq \lambda_n$.

The *spectrum* of X consists of its eigenvalues, including multiplicities.

Theorem 1.1. *Let $X = (V, E)$ be a graph without loops, with adjacency matrix A. For any $x, y \in V$ and any natural number k, the entry $A^k(x, y)$ equals the number of (x, y)-walks of length k.*

Proof. Denote by $w_k(x, y)$ the number of (x, y)-walks of length k. We will show that $A^k(x, y) = w_k(x, y)$ for any $x, y \in V$ and any natural number k using induction on k.

The base case $k = 1$ is clear from the definition of A. Let $k \geq 2$ and assume that $A^{k-1}(x, y) = w_{k-1}(x, y)$ for any $x, y \in V$. Let x and y be two vertices of X. A (x, y)-walk of length k is composed of a (x, z)-walk of length $k - 1$ followed by an edge zy. Hence,

$$w_k(x, y) = \sum_{z : z \sim y} w_{k-1}(x, z).$$

At the same time,

$$A^k(x, y) = \sum_{z \in V} A^{k-1}(x, z) A(z, y) = \sum_{z : z \sim y} A^{k-1}(x, z).$$

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From the induction hypothesis, we know that $A^{k-1}(x,z) = w_{k-1}(x,z)$ for each $z \sim y$. Using the above two equations, we get that

$$A^k(x,y) = w_k(x,y).$$

□

Now the above thereom gives the next Corollary 1.1 directly.

Corollary 1.1. *Let $X = (V, E)$ be a graph with adjacency matrix A whose eigenvalues are $\lambda_1 \geq \cdots \geq \lambda_n$. For any natural number k, the number of closed walks of length k in X equals $\sum_{j=1}^{n} \lambda_j^k$.*

Now we shall see a very interesting upper bound on the eigenvalues of a simple graph through the following proposition.

Proposition 1.1. *Let X be a simple graph with n vertices and e edges. If λ is an eigenvalue of the adjacency matrix A of X, then*

$$|\lambda| \leq \sqrt{2e\left(1 - \frac{1}{n}\right)}.$$

Proof. We use the facts that $\operatorname{tr}(A) = 0$ and $\operatorname{tr}(A^2) = 2e$. Since A is real symmetric, all its eigenvalues are real. Let the eigenvalues of A be $\lambda_1, \ldots, \lambda_n$, and let λ be one of them. The desired inequality is equivalent to

$$\lambda^2 \leq \frac{2e(n-1)}{n} \quad \Longleftrightarrow \quad \frac{n}{n-1}\lambda^2 \leq \operatorname{tr}(A^2) = \sum_{i=1}^{n} \lambda_i^2.$$

Rewriting, this is equivalent to

$$\sum_{i \neq k} \lambda_i^2 \geq \frac{1}{n-1}\lambda^2,$$

where $\lambda = \lambda_k$. Now, since $\operatorname{tr}(A) = 0$, we have, $\sum_{i \neq k} \lambda_i = -\lambda$. Applying the Cauchy–Schwarz inequality,

$$\sum_{i \neq k} \lambda_i^2 \geq \frac{1}{n-1}\left(\sum_{i \neq k} \lambda_i\right)^2 = \frac{1}{n-1}\lambda^2.$$

This proves the claim. □

2. Bipartite Graphs

A graph $X = (V, E)$ is called *bipartite* if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge of X has one endpoint in V_1 and the other in V_2.

Theorem 2.1. *If X is bipartite, and $\lambda > 0$ is an eigenvalue of the adjacency matrix A of X with multiplicity m, then $-\lambda$ is also an eigenvalue with multiplicity m.*

Proof. Assume that X is a bipartite graph whose partite sets have sizes r and s, respectively. Label the vertices so that the adjacency matrix A has the form

$$A = \begin{pmatrix} 0_r & B \\ B^T & 0_s \end{pmatrix},$$

where B is an $r \times s$ matrix.

Let $\lambda > 0$ be an eigenvalue of A with eigenvector

$$w = \begin{pmatrix} u \\ v \end{pmatrix},$$

where $u \in \mathbb{R}^r$ and $v \in \mathbb{R}^s$, not both zero. Then

$$Aw = \lambda w \implies \begin{cases} \lambda u = Bv, \\ \lambda v = B^T u. \end{cases}$$

Define

$$w' = \begin{pmatrix} u \\ -v \end{pmatrix}.$$

Then

$$Aw' = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} u \\ -v \end{pmatrix} = \begin{pmatrix} -Bv \\ B^T u \end{pmatrix} = \begin{pmatrix} -\lambda u \\ \lambda v \end{pmatrix} = (-\lambda) w'.$$

Since $w \neq 0$, we also have $w' \neq 0$. Hence $-\lambda$ is an eigenvalue of A.

A collection of m linearly independent eigenvectors corresponding to λ gives rise to m linearly independent eigenvectors corresponding to $-\lambda$. Thus the multiplicity of $-\lambda$ is at least that of λ. Repeating the argument with $-\lambda$ in place of λ shows that the multiplicities are equal. $\square$

References

[RB] R. B. Bapat, *Graphs and Matrices*, Universitext, Springer, London, 2010.

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